

JUnit Basic Testing Exercises

Exercise 1: Setting Up JUnit

Scenario:

You need to set up JUnit in your Java project to start writing unit tests.

Steps:

1. Create a new Java project in your IDE (e.g., IntelliJ IDEA, Eclipse).
2. Add JUnit dependency to your project. If you are using Maven, add the following to your pom.xml:

```
<dependency>
```

```
    <groupId>junit</groupId>
```

```
    <artifactId>junit</artifactId>
```

```
    <version>4.13.2</version>
```

```
    <scope>test</scope>
```

```
</dependency>
```

3. Create a new test class in your project.

Exercise 3: Assertions in JUnit

Scenario:

You need to use different assertions in JUnit to validate your test results.

Steps:

1. Write tests using various JUnit assertions.

Solution Code:

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```
import org.junit.Test;

import static org.junit.Assert.*;

public class AssertionsTest {

    @Test

    public void testAssertions() {

        // Assert equals

        assertEquals(5, 2 + 3);

        // Assert true

        assertTrue(5 > 3);

        // Assert false

        assertFalse(5 < 3);

        // Assert null

        assertNull(null);

        // Assert not null

        assertNotNull(new Object());

    }

}
```

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Scenario:

You need to organize your tests using the Arrange-Act-Assert (AAA) pattern and use setup and teardown methods.

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Steps:

1. Write tests using the AAA pattern.
2. Use `@Before` and `@After` annotations for setup and teardown methods.

Solution Code:

```
import org.junit.Before;

import org.junit.After;

import org.junit.Test;

import static org.junit.Assert.assertEquals;


public class AAATest {

    private int value;


    @Before

    public void setUp() {

        // Setup Method

        value = 10;

        System.out.println("Setting up test environment...");

    }


    @Test

    public void testAAA() {

        // Arrange

        int expected = 15;
```

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```
int increment = 5;
```

```
// Act
```

```
int actual = value + increment;
```

```
// Assert
```

```
assertEquals(expected, actual);
```

```
}
```

```
@After
```

```
public void tearDown() {
```

```
    // Teardown Method
```

```
    value = 0;
```

```
    System.out.println("Tearing down test environment...");
```

```
}
```

```
}
```